

# Non-Unitary Near-Groups via Endomorphisms

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- 2 Endomorphisms of Cuntz Algebras
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# What are Fusion Categories?

# Motivation I

In the 1920s, it was proposed by Dirac that with the elimination of only one basic assumption, quantum mechanics would follow naturally from the classical theory. This assumption was [commutativity](#) ([Dir26]).

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From this perspective, fusion categories can be viewed as a [quantum analogue to the finite-dimensional representation theory of finite groups](#).

This can be made quite precise as follows. For simplicity, we will always assume to be working over an [algebraically closed field of characteristic 0](#).

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By Tannakian duality ([Del90], [Del02]), every category satisfying these properties is equivalent (as a tensor category) to some  $\text{Rep}(G)$ !

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Fusion categories encompass the more general theory, with applications to operator algebras and categorical representation theory.

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- **Quadratic extensions:** “quadratic extensions” of  $\text{Vec}_G$ .
- **Extended Haagerup:** four Morita equivalent categories not known to arise from the previous three sources.

Every **known** fusion category arises from these families under elementary constructions (such as Deligne tensor products, (de-)equivariantization, Morita equivalence and graded extensions).

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In this talk, I would like to address the problem of classifying **near-group fusion categories**, beginning with Izumi's work in the unitary setting.

# Endomorphisms of Cuntz Algebras

An important result of Hayashi and Yamagami showed that **every unitary fusion category** embeds into  $\text{End}(M)$ , for  $M$  a hyperfinite type  $\text{III}_1$  factor ([HY00]). Popa's famous classification theorem for subfactors ([Pop95]) then provides a compatible notion of equivalence.

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In [Izu93] and [Izu01], Izumi showed that unitary near-group categories realized in  $\text{End}(M)$ , for  $M$  a hyperfinite type  $\text{III}_1$  factor, could be reconstructed from [Cuntz algebra endomorphisms](#).



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A **Cuntz algebra** is a universal  $C^*$ -algebra  $\mathcal{O}_n$  generated by a set of isometries  $\{S_i\}_{i=1}^n$  satisfying

$$\sum_{i=1}^n S_i S_i^* = 1.$$

Since this implies  $S_i^* S_j = \delta_{i,j}$  automatically, it follows that Cuntz algebras capture exactly the notion of **direct sums** in the unitary setting.

By [HY00], given a **type  $(G, m)$  near-group**  $\mathcal{C}$ , there are  $\rho \in \text{End}(M)$  and  $\alpha : G \rightarrow \text{Aut}(M)$  satisfying our fusion rules:

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$$X \otimes X \cong \bigoplus_{g \in G} g \oplus X^{\oplus m} \quad \rightsquigarrow \quad \rho(\rho(x)) = \sum_{g \in G} S_g \alpha_g(x) S_g^* + \sum_{i=1}^m T_i \rho(x) T_i^*.$$

These  $S_g$  and  $T_i$  are the inclusion maps for the summands  $\alpha_g$  and  $\rho$  inside  $\rho \otimes \rho$ , and hence naturally generate a Cuntz subalgebra  $\mathcal{O}_{|G|+m}$  of  $M$ .

In [Izu17], Izumi showed that  $\rho$  and  $\alpha_g$  are **polynomials in the Cuntz algebra generators**, and moreover that the  **$6j$ -symbols** of  $\mathcal{C}$  are completely determined by

- a finite-dimensional Hilbert space  $\mathcal{K} := \text{Hom}(\rho, \rho^2)$ ,
- two antilinear isometries  $j_1, j_2 \in \text{End}(\mathcal{K})$ ,
- two unitary representations  $U_{\mathcal{K}}, V : G \rightarrow \text{End}(\mathcal{K})$ ,
- a symmetric bicharacter  $\chi : G \times G \rightarrow \mathbb{T}$ ,
- a linear map  $l : \mathcal{K} \rightarrow \mathcal{B}(\mathcal{K}, \mathcal{K} \otimes \mathcal{K})$ .

He then deduced necessary and sufficient conditions for this data to reproduce  $\mathcal{C}$ , as well as a compatible notion of equivalence.

The end result of this approach was a system of polynomial equations such that there exists a **bijjective correspondence** between equivalence classes of solutions and equivalence classes of unitary near-group categories.

This classification also simplifies significantly in certain circumstances. In particular, there are strong restrictions on what  $m$  can be.

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If  $\dim(X)$  is **irrational**, then  $G$  is Abelian and  $m = k|G|$  for  $k \in \mathbb{N}$ . When  $k = 1$ , the data needed for the classification reduces to just

- two functions  $a, b : G \rightarrow \mathbb{T}$ ,
- a scalar  $c \in \mathbb{T}$

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Let's see a couple of very simple examples.

## Example

When  $G = \{0\}$ ,

- $a$  must be given by  $a(0) = 1$ ;
- $b$  must be given by  $b(0) = -1/\dim(X)$ ;
- $c$  must be given by  $c = 1$ .

Thus there is one unitary type  $(\{0\}, 1)$  near-group. This is the **Fibonacci category** and corresponds to the even part of the type  $A_4$  subfactor.

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When  $G = \mathbb{Z}/2\mathbb{Z}$ ,

- $a$  must be given by  $a(0) = 1$  and  $a(1) = \pm i$ ;
- $b$  must be given by  $b(0) = -1/\dim(X)$  and  $b(1) = (1 - a(1))/2$ ;
- $c$  must be given by  $c = \frac{1 - \sqrt{3} + a(1)(1 + \sqrt{3})}{2\sqrt{2}}$ .

Thus there are two unitary type  $(\mathbb{Z}/2\mathbb{Z}, 2)$  near-groups. These are adjoint subcategories of exceptional quantum subgroups of Temperley–Lieb–Jones.

This is a significant improvement over naïvely determining the  $6j$ -symbols!

For instance, the space  $\text{Hom}_{\mathcal{C}}((X \otimes X) \otimes X, X \otimes (X \otimes X))$  alone has dimension

$$m^4 + 3m^2|G| + |G|^2.$$

Meanwhile, in the  $m = |G|$  case, the data needed for Izumi's classification is only  $2m + 1$ .

There is also an interesting observation we can make about these two examples. The near-group fusion rules imply that

$$\dim(X) = \frac{m \pm \sqrt{m^2 + 4n}}{2}.$$

In the unitary setting, this must be positive, so there is only really one choice. In the non-unitary setting, this restriction is lifted.

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If we work with the Galois conjugates of the dimension, we will find that there *are* choices for  $(a, b, c)$  that satisfy all but two of our polynomial equations, and these correspond to non-unitary near-groups!

This is the first piece of evidence that we could expect to find a non-unitary version of Izumi's classification.

The second piece of evidence comes from Evans and Gannon.

Izumi's approach of realizing unitary fusion categories via Cuntz algebra endomorphisms, together with the work of Phillips on non-unitary analogues of Cuntz algebras ([Phi12]), inspired Evans and Gannon to devise a non-unitary version of this classification for [Haagerup–Izumi fusion categories](#) ([EG17]).

In particular, they were able to realize certain non-unitary Haagerup–Izumi categories via [Leavitt algebra endomorphisms](#).



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We would like to see if this approach can give us new examples of non-unitary near-groups. In general, the non-unitary setting is much more mysterious than the unitary setting!

# The Non-Unitary Setting: Leavitt Algebras

For us, a **Leavitt algebra** will be a universal  $\mathbb{k}$ -algebra  $\mathcal{L}_n$  with generators  $\{s_i\}_{i=1}^n \sqcup \{s'_i\}_{i=1}^n$  satisfying

$$\sum_{i=1}^n s_i s'_i = 1 \quad \text{and} \quad s'_i s_j = \delta_{i,j}.$$

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It is **not known** whether all non-unitary fusion categories may be realized as categories of endomorphisms. In the unitary setting this result comes from subfactor theory, which we no longer have access to.

So far, what we have shown is as follows. Given a type  $(G, m)$  near-group that admits a realization via algebra endomorphisms,  $\alpha_g$  and  $\rho$  will restrict to endomorphisms of a Leavitt algebra  $\mathcal{L}_{|G|+m}$ . Moreover, the  $6j$ -symbols are completely determined by

- finite-dimensional spaces  $\mathcal{K} := \text{Hom}(\rho, \rho^2)$  and  $\mathcal{K}' := \text{Hom}(\rho^2, \rho)$ ,
- two morphisms  $j'_1, j'_2 \in \text{Hom}(\mathcal{K}, \mathcal{K}')$ ,
- two automorphisms  $\varphi \in \text{Aut}(\mathcal{K})$  and  $\varphi' \in \text{Aut}(\mathcal{K}')$ ,
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- a symmetric bicharacter  $\chi : G \times G \rightarrow \mathbb{k}^\times$ ,
- two linear maps  $l : \mathcal{K} \rightarrow \mathcal{K}^2 \mathcal{K}'$  and  $l' : \mathcal{K}' \rightarrow \mathcal{K}(\mathcal{K}')^2$ .

The group-theoretic component of the project has essentially been completed.

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The goal moving forward is to focus on the case when  $\dim(X)$  is irrational, beginning with  $m = |G|$ . Izumi's unitary classification allows us to make some educated guesses as to what the constraints on our data could look like.

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The main limitation is, of course, that there could perhaps be non-unitary near-groups that escape this kind of realization. I'm very curious about results in this direction!



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Thank you very much for your attention!